

InputEvent

Inherits: [Resource](#) < [RefCounted](#) < [Object](#)

Inherited By: [InputEventAction](#), [InputEventFromWindow](#), [InputEventJoypadButton](#), [InputEventJoypadMotion](#), [InputEventMIDI](#), [InputEventShortcut](#)

Abstract base class for input events.

Description

Abstract base class of all types of input events. See [Node._input\(\)](#).

Tutorials

- [Using InputEvent](#)
- [Viewport and canvas transforms](#)
- [2D Dodge The Creeps Demo](#)
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Properties

int	device	0
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Methods

bool	accumulate (with_event: InputEvent)
String	as_text() const
float	get_action_strength (action: StringName , exact_match: bool = false) const
bool	is_action (action: StringName , exact_match: bool = false) const

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bool	is_action_pressed(action: StringName , allow_echo: bool = false, exact_match: bool = false) const
bool	is_action_released(action: StringName , exact_match: bool = false) const
bool	is_action_type() const
bool	is_canceled() const
bool	is_echo() const
bool	is_match(event: InputEvent , exact_match: bool = true) const
bool	is_pressed() const
bool	is_released() const
InputEvent	xformed_by(xform: Transform2D , local_ofs: Vector2 = Vector2 (0, 0)) const

Constants

DEVICE_ID_EMULATION = -1

Device ID used for emulated mouse input from a touchscreen, or for emulated touch input from a mouse. This can be used to distinguish emulated mouse input from physical mouse input, or emulated touch input from physical touch input.

DEVICE_ID_KEYBOARD = 16

Device ID used for input from a keyboard. This can be used to distinguish keyboard input events from joystick input events.

DEVICE_ID_MOUSE = 32

Device ID used for input from a mouse. This can be used to distinguish mouse input events from joystick input events.

Property Descriptions

int device = 0

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- [void](#) set_device(value: [int](#))

- `int` `get_device()`

The event's device ID.

Note: `device` can be negative for special use cases that don't refer to devices physically present on the system. See [DEVICE_ID_EMULATION](#).

Method Descriptions

`bool` `accumulate(with_event: InputEvent)`

Returns `true` if the given input event and this input event can be added together (only for events of type [InputEventMouseMotion](#)).

The given input event's position, global position and speed will be copied. The resulting `relative` is a sum of both events. Both events' modifiers have to be identical.

`String` `as_text()` `const`

Returns a [String](#) representation of the event.

`float` `get_action_strength(action: StringName, exact_match: bool = false)` `const`

Returns a value between 0.0 and 1.0 depending on the given actions' state. Useful for getting the value of events of type [InputEventJoypadMotion](#).

If `exact_match` is `false`, it ignores additional input modifiers for [InputEventKey](#) and [InputEventMouseButton](#) events, and the direction for [InputEventJoypadMotion](#) events.

`bool` `is_action(action: StringName, exact_match: bool = false)` `const`

Returns `true` if this input event matches a pre-defined action

If `exact_match` is `false`, it ignores additional input modifiers for `InputEventKey` and `InputEventMouseButton` events, and the direction for `InputEventJoypadMotion` events.

```
bool is_action_pressed(action: StringName, allow_echo: bool = false, exact_match: bool = false) const
```

Returns `true` if the given action matches this event and is being pressed (and is not an echo event for `InputEventKey` events, unless `allow_echo` is `true`). Not relevant for events of type `InputEventMouseMotion` or `InputEventScreenDrag`.

If `exact_match` is `false`, it ignores additional input modifiers for `InputEventKey` and `InputEventMouseButton` events, and the direction for `InputEventJoypadMotion` events.

Note: Due to keyboard ghosting, `is_action_pressed()` may return `false` even if one of the action's keys is pressed. See [Input examples](#) in the documentation for more information.

```
bool is_action_released(action: StringName, exact_match: bool = false) const
```

Returns `true` if the given action matches this event and is released (i.e. not pressed). Not relevant for events of type `InputEventMouseMotion` or `InputEventScreenDrag`.

If `exact_match` is `false`, it ignores additional input modifiers for `InputEventKey` and `InputEventMouseButton` events, and the direction for `InputEventJoypadMotion` events.

```
bool is_action_type() const
```

Returns `true` if this input event's type is one that can be assigned to an input action: `InputEventKey`, `InputEventMouseButton`, `InputEventJoypadButton`, `InputEventJoypadMotion`, `InputEventAction`. Returns `false` for all other input event types.

```
bool is_canceled() const
```

Returns `true` if this input event has been canceled.

`bool is_echo() const`

Returns `true` if this input event is an echo event (only for events of type `InputEventKey`). An echo event is a repeated key event sent when the user is holding down the key. Any other event type returns `false`.

Note: The rate at which echo events are sent is typically around 20 events per second (after holding down the key for roughly half a second). However, the key repeat delay/speed can be changed by the user or disabled entirely in the operating system settings. To ensure your project works correctly on all configurations, do not assume the user has a specific key repeat configuration in your project's behavior.

`bool is_match(event: InputEvent, exact_match: bool = true) const`

Returns `true` if the specified `event` matches this event. Only valid for action events, which include key (`InputEventKey`), button (`InputEventMouseButton` or `InputEventJoypadButton`), axis `InputEventJoypadMotion`, and action (`InputEventAction`) events.

If `exact_match` is `false`, the check ignores additional input modifiers for `InputEventKey` and `InputEventMouseButton` events, and the direction for `InputEventJoypadMotion` events.

Note: This method only considers the event configuration (such as the keyboard key or the joypad axis), not state information like `is_pressed()`, `is_released()`, `is_echo()`, or `is_canceled()`.

`bool is_pressed() const`

Returns `true` if this input event is pressed. Not relevant for events of type `InputEventMouseMotion` or `InputEventScreenDrag`.

Note: Due to keyboard ghosting, `is_pressed()` may return `false` even if one of the action's keys is pressed. See [Input examples](#) in the documentation for more information.

`bool is_released() const`

Returns `true` if this input event is released. Not relevant for events of type `InputEventMouseMotion` or `InputEventScreenDrag`.

`InputEvent xformed_by(xform: Transform2D, local_ofs: Vector2 = Vector2(0, 0)) const`

Returns a copy of the given input event which has been offset by `local_ofs` and transformed by `xform`. Relevant for events of type `InputEventMouseButton`, `InputEventMouseMotion`, `InputEventScreenTouch`, `InputEventScreenDrag`, `InputEventMagnifyGesture` and `InputEventPanGesture`.

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